Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Sunderland, Peter Ph.D.		
Project Title:	Investigation of Velocity and Temperature in Microgravity Laminar Jet Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	02/01/1999	**End Date:**	02/08/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Greenberg, Paul (NASA Glenn Research Center) Urban, David (NASA Glenn Research Center) Wernet, Mark (NASA Glenn Research Center)		
Grant/Contract No.:	None		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	This project concerns the development of two diagnostics for application to microgravity combustion experiments. These diagnostics are particle image velocimetry (PIV) and two-wavelength thin-filament pyrometry (TFP). The diagnostics will be used to measure velocities and temperatures in laminar gas jet diffusion flames in the GRC 2.2-Second Drop Tower. 1. Measure the velocity fields in microgravity laminar jet diffusion flames using particle-image velocimetry in the 2.2-Second Drop Tower. 2. Conduct full-field temperature measurements of the same flames using two-wavelength thin-filament pyrometry. 3. Compare the measurements with predictions from analytical and numerical models. These measurements will support the evaluation of analytical and numerical combustion models. Such models furnish velocity predictions, which are crucial to the analysis of past flight investigations of both laminar and turbulent jet diffusion flames.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 10/08/2025)